

"Multivariable Calculus" Day: **M T W T F S**
Date: _____
↓ Deals with multiple inputs having single output.

Multivariable Calculus:-

more than one variable in calculus.

* $y = f(x) = 2x^2 + 4x + 3$

$x \rightarrow$ Input / Independent variable

$y \rightarrow$ Output / Dependent variable

MV Calculus:-

$f(x, y)$ or $f(x, y, z)$

* $z = f(x, y) = 2x + 3y \rightarrow$ 2 inputs, single output

$x, y \rightarrow$ Input

$z \rightarrow$ Output

* $z = f(u, v, w, x) = 3u + 5v - 3w + 9x$

$u, v, w, x \rightarrow$ Input

$z \rightarrow$ Output

Suppose :- $F = ma$

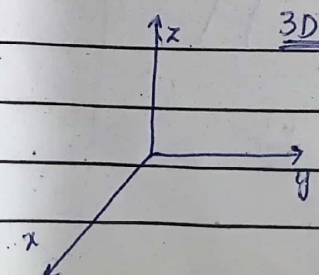
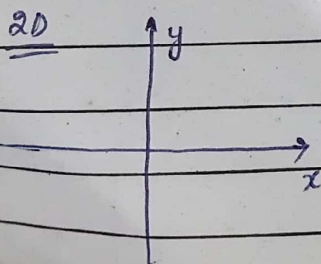
↓ ↓
single output two inputs

$\Rightarrow$ Derivative in mv called Partial Derivative.

Graph Plotting:-

• single variable $\rightarrow$ graph plotting $\rightarrow$ 2D (plane)

• multivariable $\rightarrow f(x, y) = x^2 + y^2 \rightarrow$ 3D (space)



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Day: MTWTFS

Functions of two or more variables:-

① $f(x, y) = 3x + 4y - 5$ $z = ?$ at
when $(x = 1, 2, 3, 4)$
 $(y = 2, 4, 6, 8)$.

Solution:-

$$z = 3x + 4y - 5$$

$(x, y) \rightarrow$ Independent variables

$(z) \rightarrow$ Dependent variable

$$z = f(x, y) = 3x + 4y - 5$$

$$\begin{aligned} z = f(1, 2) &= 3(1) + 4(2) - 5 \\ &= 3 + 8 - 5 \\ &= 6 \end{aligned}$$

$$\begin{aligned} z = f(2, 4) &= 3(2) + 4(4) - 5 \\ &= 6 + 16 - 5 \\ &= 17 \end{aligned}$$

$$\begin{aligned} z = f(3, 6) &= 3(3) + 4(6) - 5 \\ &= 9 + 24 - 5 \\ &= 28 \end{aligned}$$

$$\begin{aligned} z = f(4, 8) &= 3(4) + 4(8) - 5 \\ &= 12 + 32 - 5 \\ &= 39 \end{aligned}$$

- Inputs : $x = 1, y = 2$; output $z = 6$
- Inputs : $x = 2, y = 4$; output $z = 17$
- Inputs : $x = 3, y = 6$; output $z = 28$
- Inputs : $x = 4, y = 8$; output $z = 39$